Semester Test 1

Electricity and Electronics EBN111

27 March 2025

Assessment Information			
Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR answer sheet completion)	Open/Closed book:	Closed
Total number of pages (this page included)		18	

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Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of the OCR answer sheet 00/01.
4. Read the instructions on the OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. The final answers must be written as real numbers and not as fractions.
7. Use **at least 3 significant digits** to write real numbers.
8. Exact numbers in which decimal expansions are zero can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

INSTRUCTIONS

STEP 1 (-5 min to 0 min): Complete the front page of OCR answer sheet 01, and also fill in your student number on additional OCR answer sheets.

STEP 2 (0 min to 90 min): Do your calculations on the question paper.

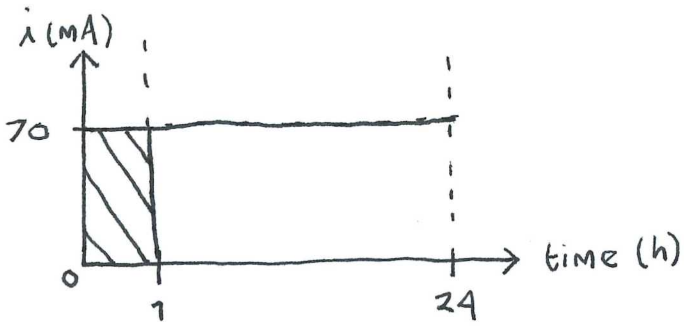
STEP 3 (90 min to 100 min): Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

- The Assessment ID is: **2025EBN111S01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should have only one character, and the characters should not touch or cross the cell.
- **Do not use commas!** Please use full stops as decimal points.

Section 1

Questions 1 to 2 [2]	
A rechargeable headlamp is capable of delivering 70 mA for 24 h.	
01	Determine the amount of charge released over a period of ONE hour. [1] 70 mC
02	If the terminal voltage is 1.5 V, how much energy (in Wh) can the battery deliver to a load over 24 h? [1] 2.52 Wh

01



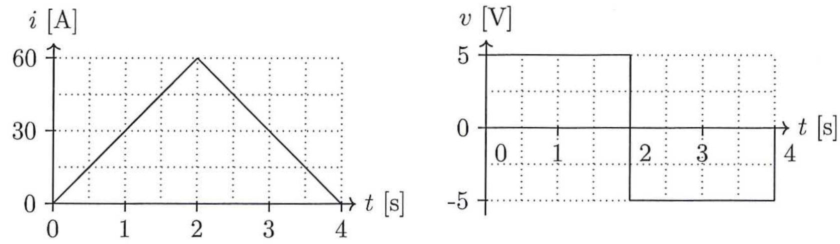
$Q = (70-0)(1-0) = 70\,mC$

02

$E = \text{Voltage} \times \text{current} \times \text{hour}$

$= 1.5 \times (70\times10^{-3}) \times 24$

$= 2.52\ \text{Wh}$

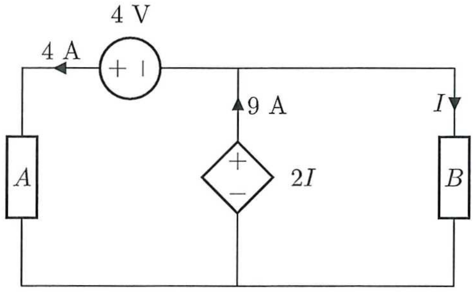
Question 3 [2]	
	
03	The current i through a circuit element and the voltage v across the element are provided above, find the total energy E for the interval $0 \leq t \leq 3$ s. [2] 525 J

$$E = \int_0^3 v(t)\, i(t)\, dt$$

$$= \left[\tfrac{1}{2}(2)(60) + \tfrac{1}{2}(1)(30) + (1)(30)\right] \times \left[(5)(2) + (-5)(1)\right]$$

$$= 525 \text{ J}$$

Questions 4 to 5 [2]



04	Calculate the power P_{4V} associated with the independent 4V voltage source. [1] $-16W$
05	Calculate the resistance R_B of circuit element B. [1] $2\,\Omega$

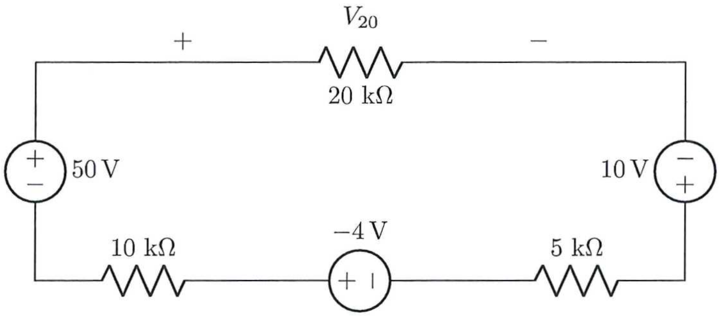
04

$$P_{4V} = (4)(-4)$$
$$= -16\text{ W}$$

05

$$R_B = \frac{2I}{I}$$
$$= 2\,\Omega$$

Question 6 [2]



06	Determine V_{20}. [2] 32 V

KVL & sign convention

$$-50 + V_{20} - 10 + V_{5k\Omega} - (-4) + V_{10k\Omega} = 0$$

$$-50 + 20\,000i - 10 + 5000i + 4 + 10\,000i = 0$$

$$i = 1.6 \times 10^{-3} \text{ A}$$

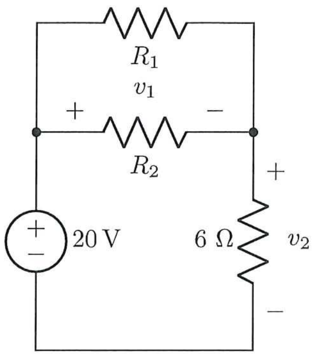
$$i = 1.6 \text{mA}$$

$$V_{20} = i \times (20\,000)$$

$$= (1.6 \times 10^{-3})(20\,000)$$

$$= \underline{32 \text{V}}$$

Questions 7 to 9 [3]
Given $R_{eq} = R_1 \| R_2 + 6 = 40\ \Omega$ and $R_1 = 2R_2$. $R_1 \| R_2$ refers to the total resistance when R_1 and R_2 are connected in parallel.



07	Determine R_1. [1]	102 Ω
08	Determine v_1. [1]	17 V
09	Determine v_2. [1]	3 V

07

$$R_{eq} = R_1 \| R_2 + 6$$
$$= \frac{R_1 R_2}{R_1 + R_2} + 6$$
$$= \frac{2R_2 \times R_2}{R_2 + 2R_2} + 6$$
$$40 = \frac{2}{3} R_2 + 6$$
$$\therefore\ R_2 = 51\ \Omega$$
$$R_1 = 2 \times 51 = \underline{102\ \Omega}$$

09

Use voltage division

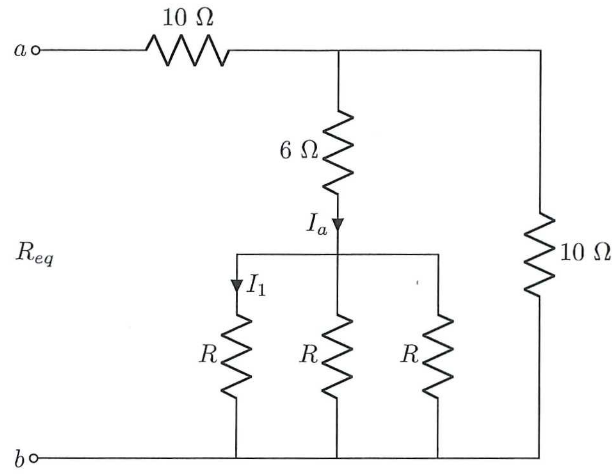
$$V_2 = \frac{6}{R_1 \| R_2 + 6} \times 20$$
$$= \frac{6}{6 + 34} \times 20$$
$$= \underline{3V}$$

08

Use KVL

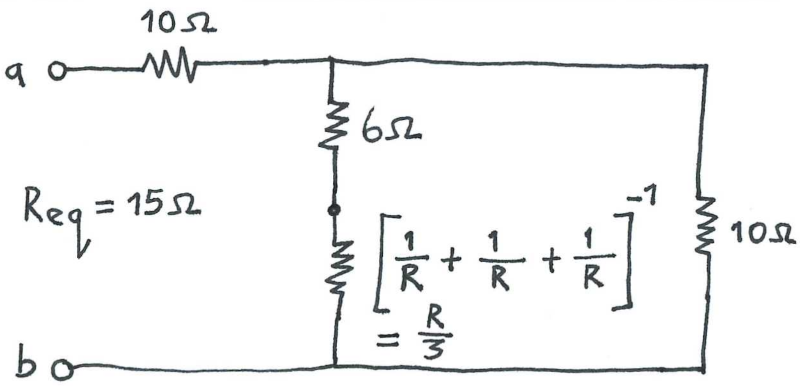
$$-20 + V_1 + V_2 = 0$$
$$-20 + V_1 + 3 = 0$$
$$V_1 = \underline{17V}$$

Questions 10 to 11 [3]



10	Determine R if $R_{eq} = 15\ \Omega$. [2] 12Ω
11	A voltage source is connected to the open terminals a-b. If $I_1 = 10$ A, determine I_a. [1] 30A

10



$$R_{eq} = 10 + \frac{(6+\frac{R}{3})(10)}{6+\frac{R}{3}+10}$$

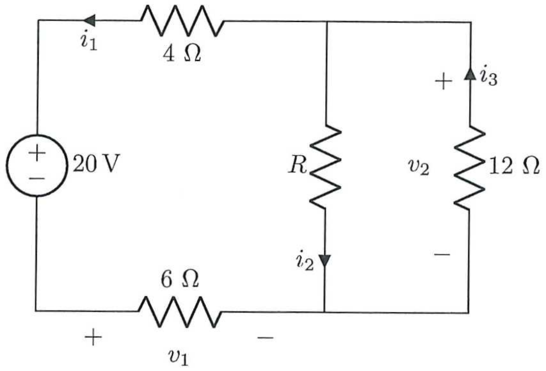
$$15 = 10 + \frac{60+\frac{10R}{3}}{16+\frac{R}{3}}$$

$R = 12\ \Omega$

11 using current division and KCL

$$I_a = I_1 + I_1 + I_1$$
$$= 3I_1$$
$$= 3(10)$$
$$= 30A$$

Questions 12 to 15 [4]



12	Determine i_1 if $R = 0\ \Omega$. [1]	−2 A
13	Determine v_1 if $R = 60\ \Omega$. [1]	−6V
14	Determine i_2 if $R = 60\ \Omega$. [1]	0.1667 A
15	Determine v_2 if $R \to \infty\ \Omega$. [1]	10.9092 V

12 KVL: $V_{4\Omega} + 20 + V_1 = 0$

$4i_1 + 20 + 6i_1 = 0$

$i_1 = -2A$

13 KVL: $V_{4\Omega} + 20 + V_1 - V_2 = 0$

$4i_1 + 20 + 6i_1 - \left[\frac{60 \times 12}{60+12} \times -i_1\right] = 0$

$i_1 = -1A$

$v_1 = 6i_1 = 6(-1) = -6V$

14 KVL: $V_{4\Omega} + 20 + V_1 - V_2 = 0$

$4(-1) + 20 + 6(-1) - V_2 = 0$

$V_2 = 10V$

$i_2 = \frac{V_2}{R} = \frac{10}{60} = 0.1667\ A$

15 KVL: $V_{4\Omega} + 20 + V_1 - V_2 = 0$

$4i_1 + 20 + 6i_1 - (12 \times -i_1) = 0$

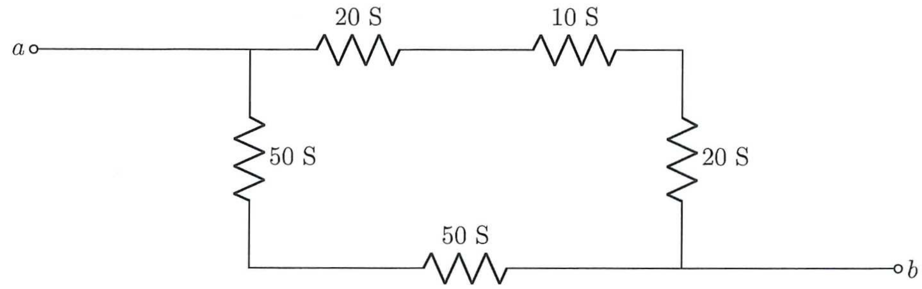
$i_1 = -0.9091\ A$

$V_2 = -12i_1$

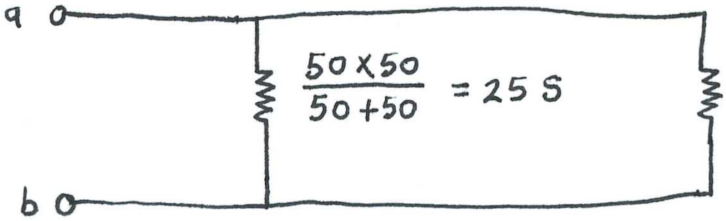
$= (-12)(-0.9091)$

$= 10.9092V$

Question 16 [2]



16	Determine the equivalent conductance G_{ab} between nodes a and b. [2] 30 S

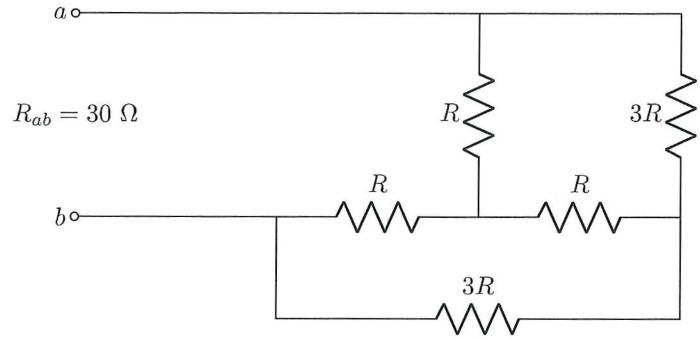


$$\frac{50 \times 50}{50 + 50} = 25\ \text{S}$$

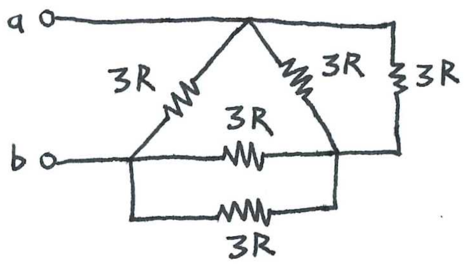
$$\left[\frac{1}{20} + \frac{1}{10} + \frac{1}{20}\right]^{-1} = 5\ \text{S}$$

$$G_{ab} = 25 + 5$$

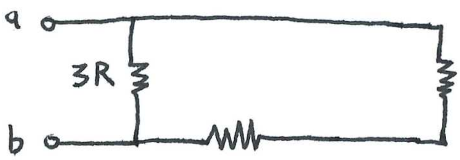
$$= \underline{30\ \text{S}}$$

Question 17 [2]	
	
17	Determine R. [2] 20Ω

Y − Δ Transformation, $R_\Delta = 3R_Y$



⇒



$$\frac{3R \times 3R}{3R + 3R} = \frac{3R}{2}$$

$$\frac{3R \times 3R}{3R + 3R} = \frac{3R}{2}$$

$$R_{ab} = 3R \parallel \left(\frac{3R}{2} + \frac{3R}{2}\right)$$

$$30 = \frac{3R}{2}$$

$$\therefore\ R = 20\Omega$$

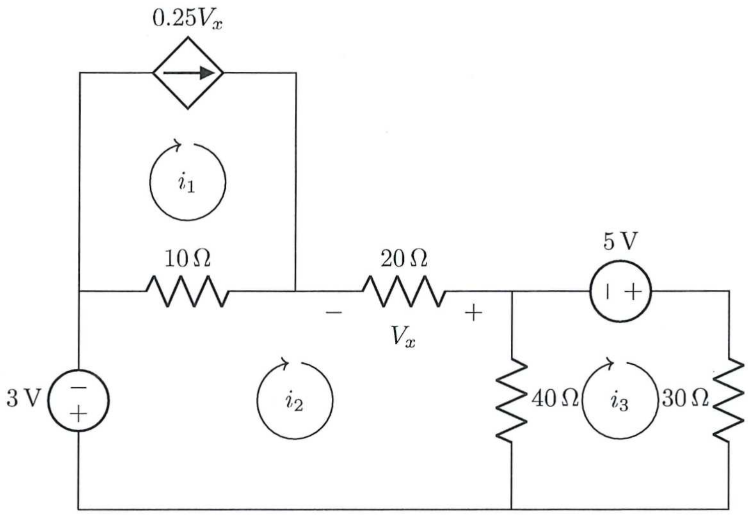
Questions 18 to 21 [4]

Use mesh analysis to set up the following two equations of the form:

$ai_2 - 40i_3 = b,$

$-40i_2 + ci_3 = d.$

Find the unknown values of a, b, c and d.



18	a [1] 120
19	b [1] −3
20	c [1] 70
21	d [1] 5

$V_x = -20i_2$

$i_1 = 0.25V_x = 0.25(-20i_2) = -5i_2$

Mesh analysis in loop 2 :

$3 + 10(i_2 - i_1) + 20i_2 + 40(i_2 - i_3) = 0$

$3 + 10i_2 - 10(-5i_2) + 20i_2 + 40i_2 - 40i_3 = 0$

$3 + 120i_2 - 40i_3 = 0$

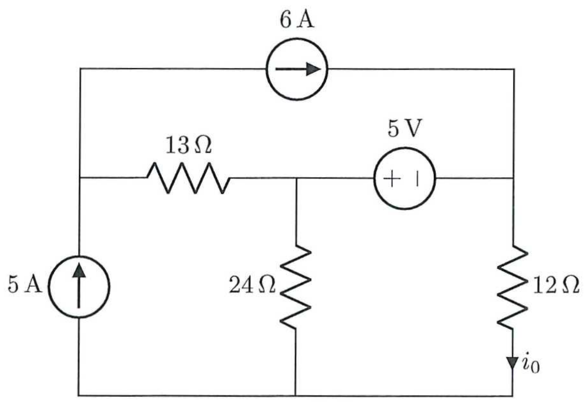
$120i_2 - 40i_3 = -3$

Mesh analysis in loop 3 :

$30i_3 + 40(i_3 - i_2) - 5 = 0$

$-40i_2 + 70i_3 = 5$

Question 22 [2]



22	Determine i_0. [2]	3.194 A

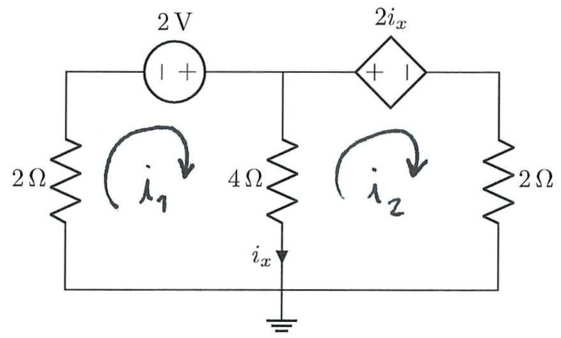
Use mesh analysis in the loop with i_0:

$$12i_0 + 24(i_0 - 5) + 5 = 0$$

$$12i_0 + 24i_0 - 120 + 5 = 0$$

$$i_0 = 3.194\ \text{A}$$

Question 23 [2]



23	Determine i_x. [2] $0.25\,A$

Using mesh analysis & KCL:

$$i_x = i_1 - i_2$$

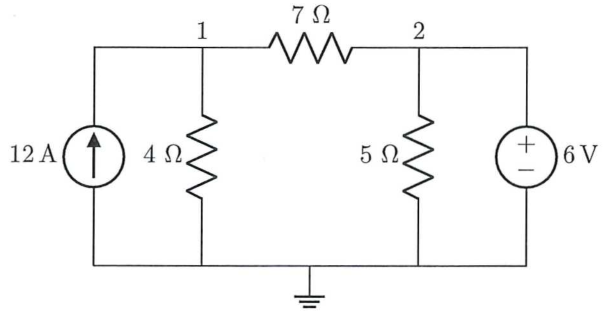
$$2i_1 - 2 + 4(i_1 - i_2) = 0$$
$$6i_1 - 4i_2 = 2$$

$$2i_2 + 4(i_2 - i_1) + 2(i_1 - i_2) = 0$$
$$-2i_1 + 4i_2 = 0$$

$$\therefore\ i_1 = 0.5 \ \&\ i_2 = 0.25$$

$$i_x = 0.5 - 0.25$$
$$= 0.25\,A$$

Question 24 [2]



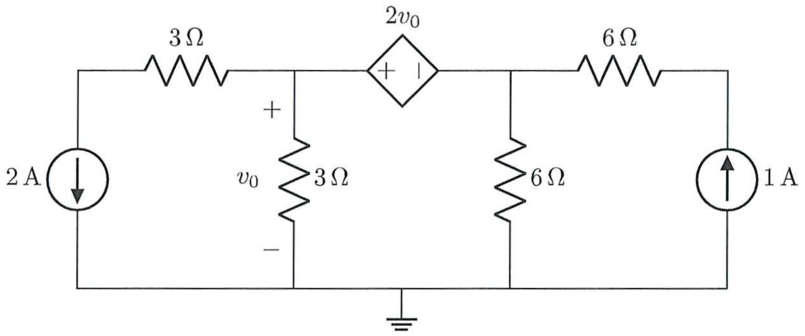
24	Solve for the node voltage v_1 at node 1. [2] 32.73 V

Use nodal analysis at node 1:

$$-12 + \frac{V_1 - 0}{4} + \frac{V_1 - 6}{7} = 0$$

$$V_1 = 32.73\ \text{V}$$

Question 25 [2]



25	Determine v_0. [2] $-6\,V$

Use nodal analysis on the supernode:

$$V_0 - V_{6\Omega} = 2V_0$$

$$V_{6\Omega} = -V_0$$

$$2 + \frac{V_0 - 0}{3} + \frac{V_{6\Omega} - 0}{6} - 1 = 0$$

$$2 + \frac{V_0}{3} + \frac{(-V_0)}{6} - 1 = 0$$

$$\therefore \underline{V_0 = -6V}$$

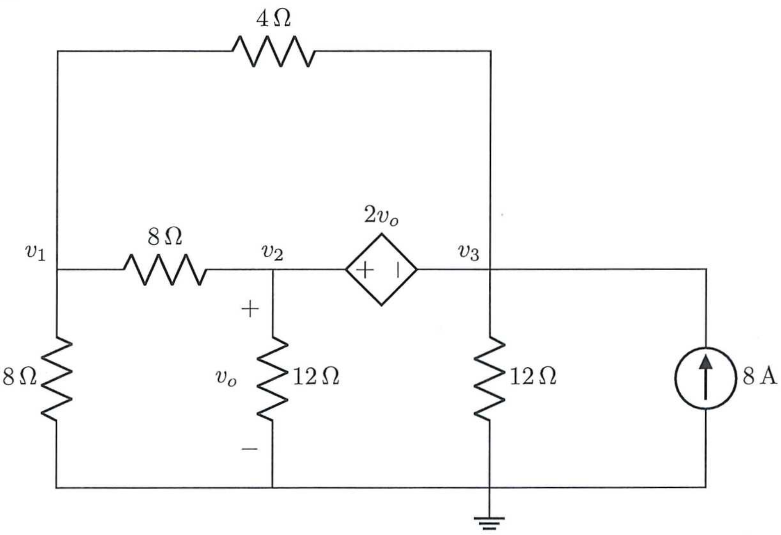
Questions 26 to 29 [4]

Use nodal analysis to set up the following two equations of the form:

$pv_1 + v_2 = q,$

$-36v_1 + rv_2 = s.$

Find the unknown values of p, q, r and s.



26	p [1]	4
27	q [1]	0
28	r [1]	−12
29	s [1]	768

For the supernode:

$$V_2 - V_3 = 2V_0$$
$$V_2 - 2V_0 - V_3 = 0$$
$$V_2 - 2V_2 - V_3 = 0$$
$$V_3 = -V_2$$

Use nodal analysis @ V_1:

$$\frac{V_1 - 0}{8} + \frac{V_1 - V_3}{4} + \frac{V_1 - V_2}{8} = 0$$
$$V_1 + 2V_1 - 2V_3 + V_1 - V_2 = 0$$
$$4V_1 - V_2 - 2V_3 = 0$$
$$4V_1 - V_2 - 2(-V_2) = 0$$
$$\underline{4V_1 + V_2 = 0}$$

Use nodal analysis around supernode:

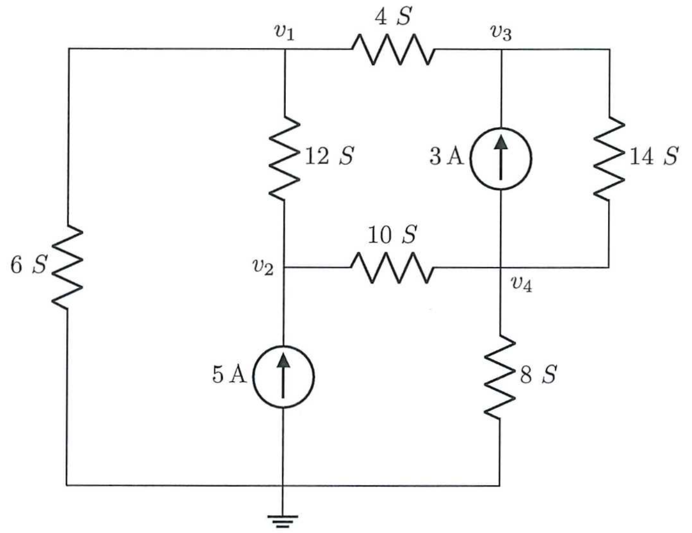
$$\frac{V_2 - 0}{12} + \frac{V_2 - V_1}{8} + \frac{V_3 - 0}{12} + \frac{V_3 - V_1}{4} - 8 = 0$$
$$2V_2 + 3V_2 - 3V_1 + 2V_3 + 6V_3 - 6V_1 - 192 = 0$$
$$-9V_1 + 5V_2 + 8V_3 = 192$$
$$-9V_1 + 5V_2 + 8(-V_2) = 192$$
$$-9V_1 - 3V_2 = 192$$
$$\underline{-36V_1 - 12V_2 = 768}$$

Questions 30 to 33 [4]

Use inspection to compile a matrix equation of the form

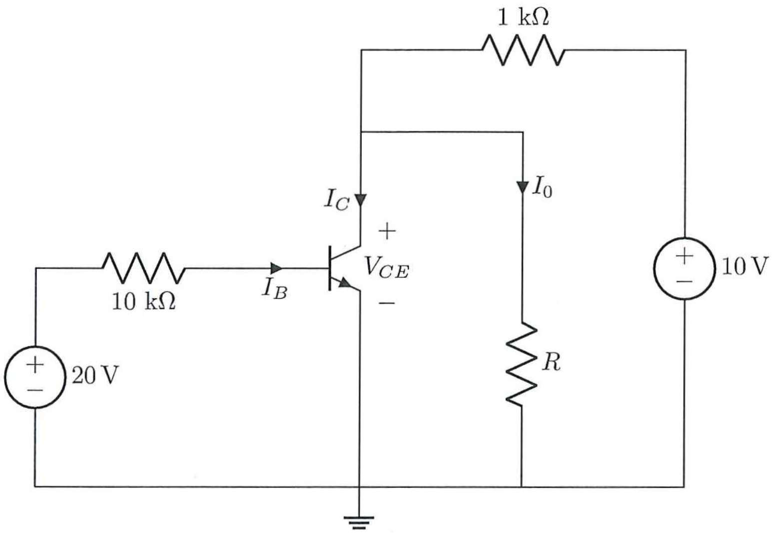
$$\begin{bmatrix} 22 & -12 & -4 & 0 \\ -12 & t & 0 & v \\ -4 & 0 & 18 & -14 \\ 0 & u & -14 & 32 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ w \\ -3 \end{bmatrix}$$

and provide the values of the matrix elements t, u, v and w.



30	t [1]	22
31	u [1]	−10
32	v [1]	−10
33	w [1]	3

$$\begin{bmatrix} 6+12+4 & -12 & -4 & 0 \\ -12 & 12+10 & 0 & -10 \\ -4 & 0 & 4+14 & -14 \\ 0 & -10 & -14 & 10+8+14 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 3 \\ -3 \end{bmatrix}$$

Questions 34 to 35 [4]		
$I_B = 15\ \mu$A and $\beta = 100$.		
		
34	Determine V_{CE} if $R \to \infty$. [2]	$8.5V$
35	Determine I_0 if $R = 5$ kΩ. [2]	$1.417mA$

$I_c = \beta I_B = (100)(15\times10^{-6}) = 1.5mA$

38 KVL for output loop:

$-V_{CE} + V_{1k\Omega} + 10 = 0$

$-V_{CE} - (1.5\times10^{-3})(1\times10^{3}) + 10 = 0$

$V_{CE} = 8.5V$

39 Nodal analysis @ V_{CE}:

$$\frac{V_{CE} - 0}{5\,000} + \frac{V_{CE} - 10}{1\,000} + 1.5\times10^{-3} = 0$$

$V_{CE} = 7.0833V$

$$I_0 = \frac{V_{CE} - 0}{R} = \frac{7.0833}{5\,000} = 1.417mA$$

Total Section 1: [46]

Grand Total : [46]